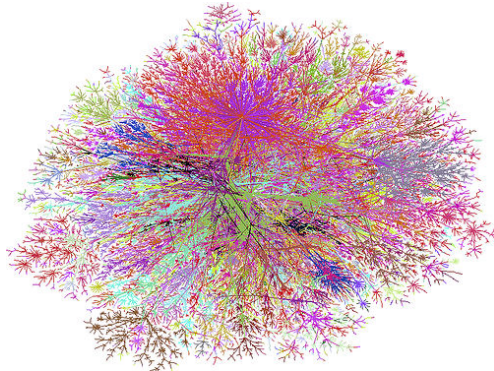


Networks

- ▶ A network is simply a collection of connected objects
- ▶ We refer to the objects as **nodes** or **vertices**, and usually draw them as points
- ▶ We refer to the connections between the nodes as **edges** or **links**, and usually draw them as lines between points
- ▶ In mathematics, networks are often referred to as **graphs**

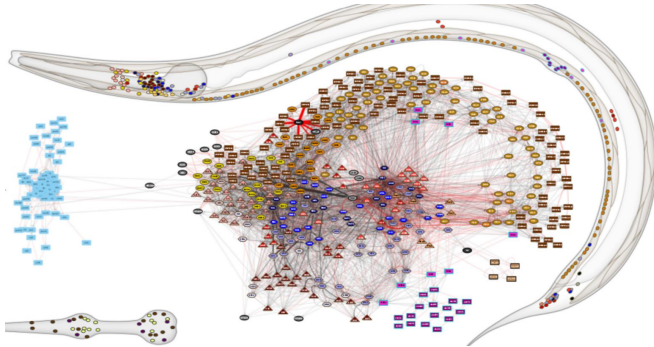
- ▶ Networks can represent all sorts of systems in the real world
- ▶ Internet: nodes are computers or other devices, and the edges are physical (or wireless, even) connections between the devices
- ▶ World Wide Web: pages are nodes and links are the edges
- ▶ Social networks of acquaintances or other types of interactions
- ▶ Networks of publications linked by citations or collaborations
- ▶ Transportation networks
- ▶ Metabolic networks

Network of connections between devices within the Internet



Courtesy of Steve Jurevson

Biological Neural Network of C Elegans



Courtesy of the OpenWorm project

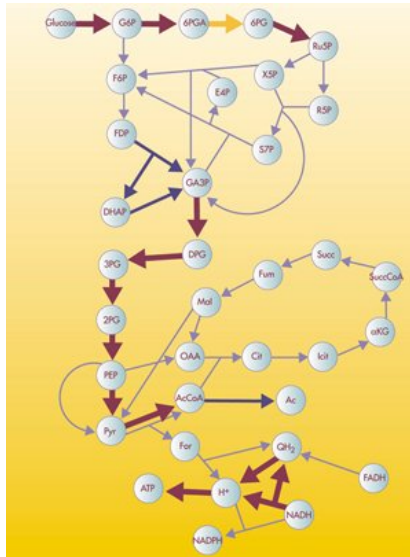
C. Elegans nematodes are microscopic roundworms with only 302 neurons, used to study basic biological mechanisms and disease.

This wiring diagram is a network of identifiable, labeled neurons connected by chemical and electrical synapses.

Metabolic network model for *Escherichia coli*

Network models of *E. Coli* are computational systems biology frameworks that map metabolic, regulatory, and transcriptional interactions to predict behavior, primarily using genome-scale metabolic models (GEMs).

These networks include nodes for genes, proteins, and metabolites, aiding in metabolic engineering, strain design, and studying stress responses.



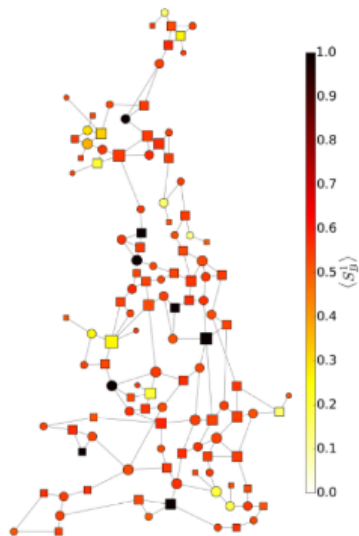
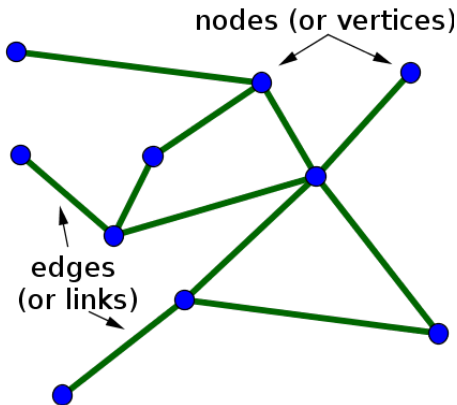


FIG. 5. Network topology of the power transmission grid of the United Kingdom (comprising $N = 120$ nodes) with second-order Kuramoto-type nodal dynamics. Circular nodes are net generators while square nodes are net consumers. The size of each node is proportional to the degree, and its color corresponds to the $\langle S_B^1 \rangle$ value of the respective node.

Undirected Networks

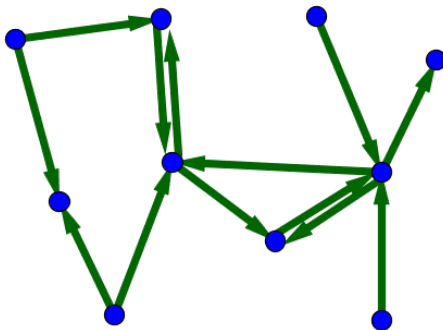
An example with 10 Nodes and 11 Edges



Directed Networks

For instance, consider, the World Wide Web : Links between pages are directed

In such a **Directed Graph** (or digraph, for short), we typically draw the edges as arrows to indicate the direction



Weighted Networks

- ▶ In some networks, not all nodes and edges are equal
- ▶ For example, in metabolic networks, nodes may indicate different enzymes which have a wide variety of behaviors, and edges may indicate vastly different types of interactions
- ▶ To model such difference, one can introduce different types of nodes and edges in the network
- ▶ In networks where the differences among nodes and edges can be captured by a single number that, for example, indicates the strength of the interaction, a good model may be a weighted graph
- ▶ One can graphically represent a weighted graph by different sizes of nodes and edges with different linewidths

For simplicity, we will focus primarily on unweighted graphs with a single type of node and a single type of edge

We will also consider undirected graphs, and not allow multiple connections or self-connections

Adjacency Matrix

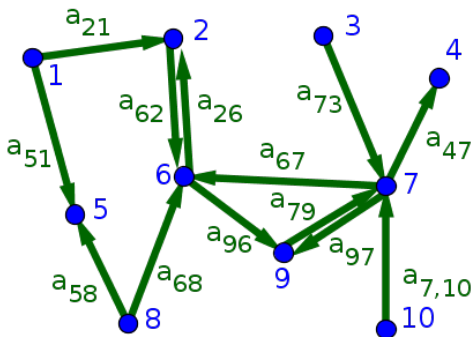
For unweighted networks of N nodes without multiple connections, the network structure can be represented an $N \times N$ adjacency matrix A

Define the adjacency matrix so that the component a_{ij} indicates a connection from node j and to node i

Adjacency matrix is a matrix of ones and zeros, defined by:

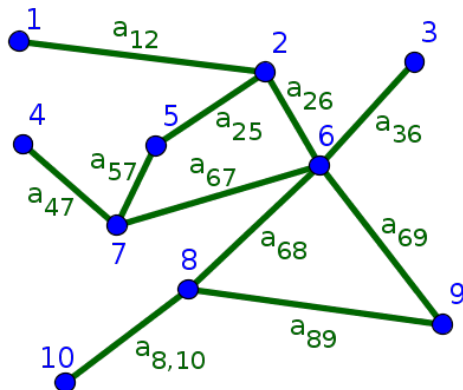
- ▶ $a_{ij} = 1$ if there is an edge from node j to node i
- ▶ $a_{ij} = 0$ if nodes j and i are unconnected

Directed Network



$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Undirected Network



For undirected graphs, all edges are bidirectional, implying that the adjacency matrix is **symmetric**

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

Degree distribution of a network

Important feature: Number of connections a node has, i.e., the degree of the node

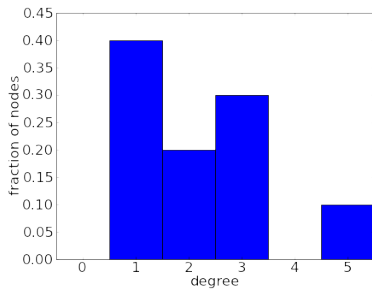
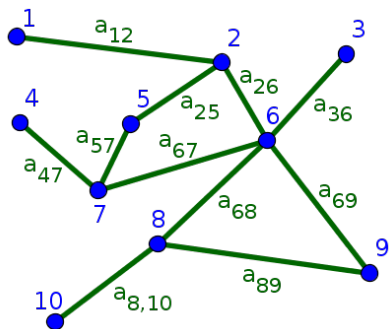
If we restrict ourselves to undirected networks for the moment, then the degree of a node i in terms of the adjacency matrix A , is :

$$k_i = \sum_j a_{ij}$$

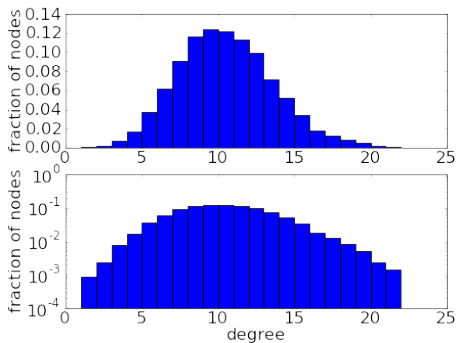
where the sum is over all nodes in the network

By counting how many nodes have each degree, we form the degree distribution $P_{deg}(k)$, defined by

$P_{deg}(k)$ = Fraction of nodes in the graph with degree k

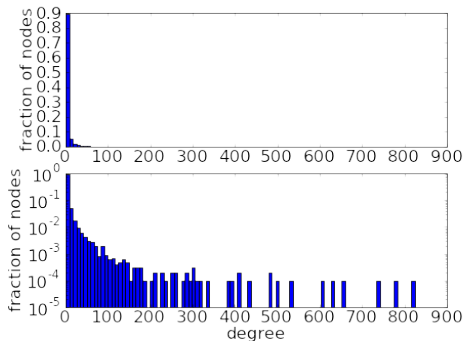


- ▶ In the simplest types of networks, one would find that most nodes in the network have similar degrees
- ▶ However, real world networks may have very different degree distributions
- ▶ In many real world network, most nodes have a relatively small degree, but a few nodes will have very large degree, being connected to many other nodes
- ▶ These large-degree nodes are often referred to as hubs, in analogy to transportation network such as one connecting airports, where some very large hub airport have connections to many others
- ▶ Such a degree distribution is said to have a long tail



Histograms of a binomial degree distribution of a network with 10,000 nodes (top on linear scale; bottom on log scale)

Probability of a connection is $p = 10/10,000$, so the average degree is 10



Histograms of a **power law degree distribution** of a network with 10,000 nodes : power law exponent of -2

Degree of a Directed Network

Add up the incoming connections and outgoing connections separately, obtaining two numbers for the degree of a node

In-degree of node i is the total number of connections onto node i , and is the sum of the i -th row of the adjacency matrix:

$$k_i^{in} = \sum_j a_{ij}$$

Out-degree of node i is the total number of connections coming from node i , i.e. sum of the i -th column of the adjacency matrix:

$$k_i^{out} = \sum_j a_{ji}$$

Total degree: $k_i^{tot} = k_i^{in} + k_i^{out}$

Mean Path Length

Let d_{ij} be the length of the shortest path (geodesic) between nodes i and j

In a network, the length of a path is the number of edges that the path contains

The mean path length is the average of the shortest path length, averaged over all pairs of nodes

For an undirected graph of N nodes, the mean path length is:

$$L = \frac{1}{N(N-1)} \sum_{i \neq j} d_{ij}$$

where the sum is over all pairs of distinct nodes

If two nodes are disconnected, meaning there is no path between them, then the path length between them is infinite

As a consequence, if a network contains disconnected components (collections of nodes that have no paths between them), then the mean path length L also diverges to infinity

One way to avoid this problem is to calculate L only from nodes in the largest connected component

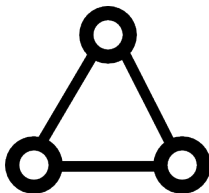
Clustering

One property of real world networks can be summarized by “**your friends are often friends**”

Think of a network describing friendship, where nodes are people and a link exists between nodes if the people are friends

Then two nodes that are connected to a third node are more likely to be connected themselves

This property increases the frequency of finding the **triangle motif** that consist of three mutually connected nodes



Definition of Clustering Coefficient

The clustering coefficient of an undirected graph is a measure of the number of triangles in a graph

The clustering coefficient of a graph is based on a local clustering coefficient for each node:

$$C_i = \frac{\text{number of triangles connected to node } i}{\text{number of triples centered around node } i}$$

where a triple centered around node i is a set of two edges connected to node i

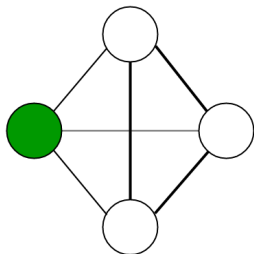
If the degree of node i is 0 or 1, which gives us $C_i = 0/0$, we can set $C_i = 0$

The clustering coefficient for the whole graph is the average of the local values C_i

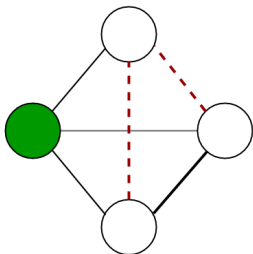
$$C = \frac{1}{n} \sum_{i=1}^n C_i$$

where n is the number of nodes in the network

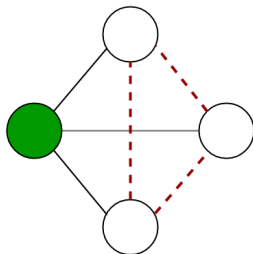
By definition $0 \leq C_i \leq 1$ and $0 \leq C \leq 1$



$C = 1$



$C = 1/3$



$C = 0$

Definition of the Transitivity of a Graph

The transitivity T of a graph is based on the relative number of triangles in the graph, compared to total number of connected triples of nodes:

$$T = \frac{3 \times \text{number of triangles in the network}}{\text{number of connected triples of nodes in the network}}$$

The factor of three in the number accounts for the fact that each triangle contributes to three different connected triples in the graph, one centered at each node of the triangle

With this definition, $0 \leq T \leq 1$, and $T = 1$ if the network contains all possible edges

The transitivity of a graph is closely related to the clustering coefficient of a graph, as both measure the relative frequency of triangles

Centrality

- ▶ Centrality is a crucial concept in graph analytics: distinguishes important nodes in a graph
- ▶ It recognizes nodes that are important or central among the whole list of other nodes in a graph.
- ▶ Different things can define the importance of every node: various metrics used from different perspectives
- ▶ Simplest of all the centrality measures of node connectivity: Degree Centrality

Closeness Centrality

- ▶ Closeness Centrality identifies a node's importance based on how close it is to all the other nodes in the graph.
- ▶ The closeness is also known as geodesic distance, which is the number of links included in the shortest path between two nodes.
- ▶ To calculate that closeness for a node, sum up all the shortest paths between that node and all the other nodes in the network on the graph.
- ▶ Closeness centrality finds application in identifying individuals that are in a position to influence the entire network in the fastest way possible.

Betweenness Centrality

- ▶ Betweenness Centrality defines the importance of any node based on the number of times it occurs in the shortest paths between other nodes.
- ▶ It measures the percent of the shortest path in a network and where a particular node lies in it.
- ▶ A node with high betweenness centrality is considered the most influential one over other nodes in the network.
- ▶ This is because this measure can provide insights into the most critical path as disrupting them will disrupt the network.

Eigenvector Centrality

- ▶ Eigenvector Centrality defines a node's importance based on the function of its neighboring nodes.
- ▶ For instance, consider a node in a network. Check all the nodes it's connected to. If a node is linked to or surrounded by highly important nodes in a network, it has a high eigenvector centrality score.
- ▶ Relationships with high scoring nodes have more contribution to the score of a node than connections to nodes with low eigenvector centrality scores. It is what makes it an important part of the whole network.
- ▶ A subvariant in the Eigenvector Centrality is PageRank: It is defined as the measure of directional influence of nodes and, thus, is most suited for directed graphs.

- ▶ **Assortative mixing** : Tendency of nodes to connect to other nodes that are alike
- ▶ **Community Structure** : Network nodes are joined together in tightly knit groups, between which there are only looser connections

Random networks

Erdős-Rényi Network (a.k.a. Poisson Network) is a random graph $G(N, p)$ with N labeled nodes where each pair of nodes is connected by a preset probability p :

- ▶ Fix node number N
- ▶ Among all possible edges, include each edge with probability p independently
- ▶ N and p do not uniquely define the network: there are $2^{\binom{N}{2}}$ different realizations of it.

The parameter p in this model can be thought of as a weighting function

Average density of links is p

As p increases from 0 to 1, the model becomes more and more likely to include graphs with more edges and less and less likely to include graphs with fewer edges

Degree Distribution for the ER Random Network is Binomial, with mean degree $\langle k \rangle = (N - 1)p$ and variance $\sigma^2 = (N - 1)p(1 - p)$

For large N and small p , this is approximately a Normal Distribution

ER random networks have a **low average path length**, meaning that there tends to be a path between a pair of nodes that involves only a few edges : presence of short-cuts

Average path distance in the random network is approximately

$$\langle L \rangle \approx \frac{\log N}{\log \langle k \rangle}$$

This property is shared by many real world networks

This notion has been popularized by terms like the “six degrees of separation” between any two people, meaning they are typically connected by a chain of six or fewer edges in a social network graph where people are viewed as nodes and acquaintances are linked by an edge in the graph

If the edges in a network are generated randomly, then the fact that two nodes have a common neighbor cannot increase the probability that the nodes are connected

For example, in the following ER random network of 50 nodes and 0.1 connection probability, each node is connected to around 5 other nodes

If you start at a node and follow the connections to two of its neighbors, those neighbors will still have only a small chance (the original 10% connection probability) of being connected

So clustering is small for ER random networks

The average Clustering Coefficient in the ER Random Network is approximately:

$$\langle C \rangle \approx \frac{\langle k \rangle}{N}$$

i.e. $\langle C \rangle \approx p$

- ▶ No Community Structure
- ▶ No Assortative Mixing

Giant Component in the ER Random Network

- ▶ The largest component in the ER random graph has constant size 1 when $p = 0$ and extensive size N when $p = 1$
- ▶ An interesting question to ask is how the transition between these two extremes occurs if we construct random graphs with gradually increasing values of p , starting at 0 and ending up at 1 (like bond percolation)
- ▶ It turns out that the size of the largest component undergoes a sudden change, or phase transition, from constant size to extensive size at $p = 1/N$

Size of Giant Component in the ER Random Network

Threshold for emergence of Giant Component is $p = \frac{1}{N}$, i.e.
 $\langle k \rangle \approx 1$

If $p < \frac{1}{N}$

- ▶ with high probability, there is no giant component, with all connected components of the graph having size $O(\log N)$

If $p > \frac{1}{N}$

- ▶ with high probability, there is a single giant component, with all other components having size $O(\log N)$

Small World Networks

Two properties of many real world networks are that the distance between any pairs of nodes is relatively small while at the same time the level of transitivity, or clustering is relatively high

For ER random networks, L is small, as commonly found in real-world networks

But clustering is small, while they are comparatively large in most real world networks

- ▶ One way to increase clustering is to arrange all the nodes in a lattice, like a ring of nodes
- ▶ If one connects each node just to nodes that are nearby, clustering in the resulting network naturally rises
- ▶ Since a pair of nodes that are connected to a third node will tend to be close to each in the lattice, it is likely they are also connected
- ▶ The number of triangles in the graph will be relatively high, resulting in high transitivity or clustering

In the ring lattice, it is known that the clustering coefficient C is

$$C = \frac{3}{4} \frac{(k-2)}{(k-1)}$$

This predicts that C becomes about $3/4$, when degree is sufficiently large

- ▶ However, such a lattice model will not have the small world property
- ▶ Since all connections are to nearby nodes, it will take a long chain of connections to reach a node that is large distance away on the lattice
- ▶ If the nodes are arranged on a circle, then the shortest path between nodes at opposites of the circle will be quite large

Constructing a Small World Network

One network model that yields a high clustering coefficient (but still keeps a small mean path length) is the **Small World Network model**

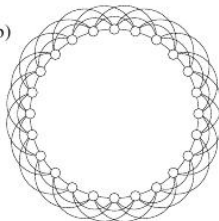
Watts and Strogatz (Nature, 1998) developed this model, combining the clustering of lattice models with the low path length of the random network model

- ▶ They start with the lattice model that has high clustering and high mean path length
- ▶ Then, they add to the model a probability p that an edge is **rewired**, meaning that the edge is disconnected from one of its nodes and then randomly connected to another node anywhere in the network
- ▶ Each edge is chosen to be rewired independent with probability p

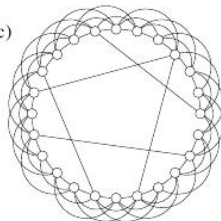
(a)



(b)



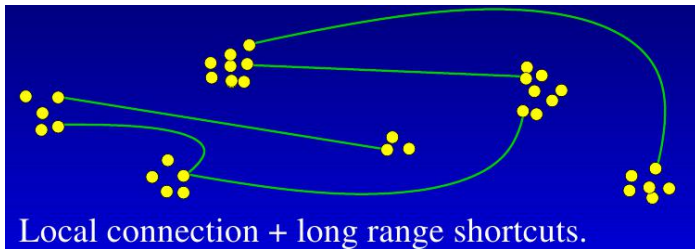
(c)



Many networks lie somewhere between these two extremes of completely regular or completely random

- ▶ Choose a vertex and the edge that connects it to its nearest neighbour
- ▶ With probability p , we reconnect this edge to a vertex chosen uniformly at random over the entire ring (duplicate edges forbidden)
- ▶ Repeat this process around the ring
- ▶ If degree is larger than 2, consider the edges that connect vertices to their second-nearest neighbours, and repeat the procedure

- ▶ When the probability p is low, then most connections are still the original local connections that they connect nodes that are nearby in the lattice
- ▶ For this reason, the clustering is still high
- ▶ However, some of the edges that have been rewired might turn into long distance connections that connect nodes that are far away from each other in the lattice
- ▶ These long distance connections, or shortcuts, immediately create a short distance between the nodes around either end of the shortcut



When the rewiring probability p becomes large, the network becomes similar to the ER random network model and the transitivity becomes low

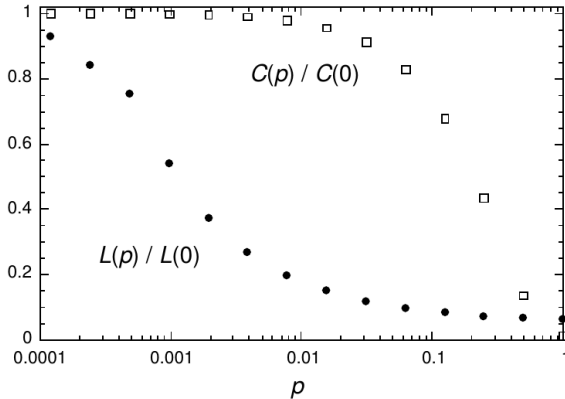
But, there is a large range of p where the clustering coefficient C is large and the average path length L is small

For this range of the parameter p , the network is considered a small world network.

Table 1 Empirical examples of small-world networks

	L_{actual}	L_{random}	C_{actual}	C_{random}
Film actors	3.65	2.99	0.79	0.00027
Power grid	18.7	12.4	0.080	0.005
<i>C. elegans</i>	2.65	2.25	0.28	0.05

Watts & Strogatz, Nature, 1998



$L(p)$ measures the typical separation between two vertices in the graph (a global property)

$C(p)$ measures the cliquishness of a typical neighbourhood (a local property)

Specifically consider networks that have many vertices with sparse connections, but not so sparse that the graph is in danger of becoming disconnected:

$$N \gg k \gg \ln(N) \gg 1$$

where $k \gg \ln N$ guarantees that a random graph will be connected

As $p \rightarrow 0$:

$$L \sim N/2k \gg 1; \quad \text{and} \quad C \sim \frac{3}{4}$$

As $p \rightarrow 1$

$$L \approx L_{\text{random}} \sim \frac{\ln(N)}{\ln(k)}; \quad \text{and} \quad C \approx C_{\text{random}} \sim \frac{k}{N}$$

- ▶ Regular lattice at $p = 0$: Highly clustered, large world where L grows linearly with N
- ▶ Random network at $p = 1$: Poorly clustered, small world where L grows only logarithmically with N
- ▶ Broad interval of p over which $L(p)$ is almost as small as L_{random} yet $C(p) \gg C_{random}$

Immediate drop in $L(p)$ caused by the introduction of a few long-range edges: short-cuts connect vertices that would otherwise be much farther apart

For small p , each short cut has a highly nonlinear effect on L , contracting the distance not just between the pair of vertices that it connects, but between their immediate neighbourhoods, neighbourhoods of neighbourhoods and so on

By contrast, an edge removed from a clustered neighbourhood to make a short cut has, at most, a linear effect on C ; hence $C(p)$ remains practically unchanged for small p (i.e. the effect of short-cuts at a local level is almost undetectable)

Scale Free Network

A scale-free network is a connected graph or network with the property that the number of links k originating from a given node exhibits a power law distribution

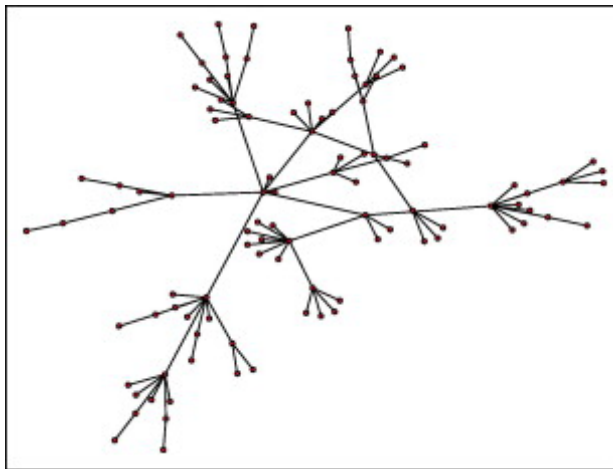
$$P(k) \sim k^{-\gamma}$$

A scale-free network can be constructed by progressively adding nodes to an existing network and introducing links to existing nodes with **preferential attachment** so that the probability of linking to a given node i is proportional to the number of existing links k_i that node has, i.e. **“rich get richer”** :

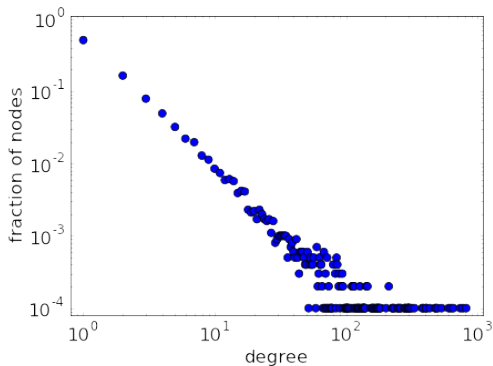
$$P(\text{linking to node } i) \sim \frac{k_i}{\sum_j k_j}$$

- ▶ Start with m_0 nodes
- ▶ At each time step, we add a node with m ($m \leq m_0$) edges connected to the nodes already in the network.
- ▶ When selecting the edges, the probability of a node being selected is proportional to the degree of that node.

Scale free network of 100 nodes with $m = 1$



Scale-free network of $N = 10,000$ nodes and power-law exponent of 2



The average degree is about 7, but 3/4 of the nodes have a degree of 3 or less ; Long tail of the degree distribution give rise to hubs

Average path length approaches

$$L \approx \frac{\log(N)}{\log(\log(N))}$$

Clustering coefficient decreases with increasing degree

Scale-free networks occur in many areas of science and engineering, for instance:

- ▶ Web pages (where the nodes are individual web pages and the links are hyper-links)
- ▶ Power grid of the western United States (where the nodes are generators, transformers, and substations and the links are power transmission lines)
- ▶ Peer-reviewed scientific literature (where the nodes are publications and the links are citations)

Barabasi AL and Albert R. *Statistical mechanics of complex networks*. Rev. Modern Physics, 7:47-97, 2002

A general form of N coupled dynamical oscillators is:

$$\dot{X}_i = F(X_i) + \sum_{j=1}^N a_{ij} G_{ij}(X_i, X_j), \quad i = 1, \dots, N$$

- ▶ X_i denotes the dynamical variables of the i th oscillator
- ▶ a_{ij} are the elements of the relevant $N \times N$ Adjacency matrix that contains information on the coupling topology
- ▶ G_i is the coupling function that represents the nature of the interaction and the variables involved in the interaction term